DILLAN IMANS

ML Research Engineer | Medical Imaging & Computer Vision

Relocating to London, UK (Oct 2026)
dillanimansbusiness@gmail.com | github.com/DillanImans | linkedin.com/in/dillanimans

SUMMARY

ML Research Engineer with first-author publications at IEEE BIBM 2025 and Diagnostics, specializing in medical image segmentation, multimodal learning, and clinical AI deployment across brain MRI, retinal imaging, and surgical vision. Incoming MSc AI for Biomedicine and Healthcare at UCL.

EDUCATION

MSc AI for Biomedicine and Healthcare, University College London — **Oct 2026 - Oct 2027 (Expected)**

B.S. Computer Science & Engineering, Sungkyunkwan University — **Aug 2022 - Jun 2026**
95.9 / 100 · Dean's List 2023 · Academic Excellence Scholarship

EXPERIENCE

AI Research Engineer · Superintelligence Lab SKKU, South Korea — **Oct 2024 - Present**

- Developed unsupervised domain adaptation for brain tumor segmentation, outperforming benchmarks by ~5-10% AUC across domain shift (first-author, IEEE BIBM 2025)
- Built clinical eye analysis systems including end-to-end fundus analysis (optic cup/disc, vessel analysis, CDR) and glaucoma detection pipeline, deployed in an industry-partnered clinical project for retinal disease screening
- Initiated cross-modality distillation framework for hypertension risk prediction from unpaired MRI, ECG, and fundus images; under review at MICCAI 2026

AI Research Engineer · Labren CUHK, Hong Kong — **Jun 2024 - Present**

- Led pipeline design, annotation QA, and preprocessing infrastructure for the first large-scale surgical action dataset (11,915 clips)
- Collaborated with NVIDIA on a multimodal surgical robotics dataset; supported annotation, preprocessing, and pipeline development
- Co-authored 3 papers targeting ECCV 2026, IEEE TMI, and npj Digital Surgery

AI Research Engineer · Infolab SKKU, South Korea — **Feb 2024 - Oct 2024**

- Published first-author paper in Diagnostics (Oct 2024): explainable multi-layer ensemble model for depression detection and severity prediction
- Designed interpretable clinical ML pipeline using behavioral features for mental health screening

PROJECTS

Computational Modeling of Collagen Triple-Helix Mutations

- Modeled COL1A1 triple-helix segments using ColabFold to investigate how Gly→Val substitutions disrupt structural stability; quantified helix distortion via RMSD analysis, linking structural defects to osteogenesis imperfecta mechanisms

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C/C++, Java, SQL, Bash
ML / AI: PyTorch, MONAI, HuggingFace, scikit-learn, OpenCV, Numpy, Pandas
Full-Stack: React/Redux, Node.js, MongoDB, Flask, Firebase, REST APIs
Tools: Git, Linux, Docker, AWS, NiBabel, SimpleITK, AlphaFold, BLAST
Spoken Languages: English & Indonesian (Fluent), Mandarin & Korean (Conversational)